

UQFF Galactic Discrete Gravity Bands & Aether Field Simulator

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PAPER_655: UQFF Galactic Discrete Gravity Bands & Aether Field Simulator ****Author:**** Daniel T. Murphy

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Companion papers: PAPER_650 (Buoyancy Harmonics), PAPER_646 (Ui Operator), PAPER_647 (Vacuum Density), PAPER_642 (SM Bridge)

Abstract

$$U_{g1} = k_1 \frac{G M \mu_B B}{r^3}; \quad U_{g2} = k_2 \frac{G M \epsilon_0 E^2}{2r}; \quad U_{g3} = k_3 \sum_j B_j \cos(\omega_s t + \pi) P_{\text{core}} E_{\text{react}}$$

The SystemAnalysisSimulator (v1-v7) implements three simultaneous, discrete Universal Gravity bands (Ug1: internal magnetic dipole, Ug2: field bubble, Ug3: magnetic string disk) operating on galactic objects within the Universal Aether ($\rho_{vac,A} = 10^{-23} \text{ gm/cm}^3$). The v7 version adds **discrete Universal Magnetism** (non-interactive, separately banded), confirms that each Ug band has a paired opposite Ub (buoyancy) band, and derives star spin rate as a function of Ug1/Ub/Ug2. The simulator uses 82 real two-star galactic observational data points for validation. This paper formalizes the three-band gravity architecture, the non-interactive magnetism condition, and the Aether density 10^{-23} gm/cm^3 as the medium supporting all field propagation.

§1 Three Discrete Universal Gravity Bands

1.1 Ug1 — Internal Magnetic Dipole Gravity

$$U_{g1} = k_1 \cdot \frac{G M \mu_B B_{\text{internal}}}{r^3} \cdot (1 + H_{SCm})$$

Variable	Value	Meaning
k_1	UQFF calibrated	Dipole coupling constant
G	$6.674 \times 10^{-11} \text{ N} \cdot \text{m}^2 / \text{kg}^2$	Newton's gravitational constant
M	body mass (kg)	Central body mass
μ_B	$9.274 \times 10^{-24} \text{ J/T}$	Bohr magneton
B_internal	body dipole field (T)	Internal dipole field strength
r	separation distance (m)	From body center
H_SCm	0.99	Heliosphere/superconductive modulation

Physical meaning: Ug1 is the gravity component sourced by the body's **internal magnetic dipole** — essentially the coupling between mass (GM), *magnetic moment* (μ), and the inverse-cube field geometry of a dipole ($1/r^3$).

1.2 Ug2 — Field Bubble Tension

$$U_{g2} = k_2 \cdot \frac{GM}{\epsilon_0} E_{\text{field}}^2 \cdot \left(\sum_j \rho_{\text{vac},j} \right) \cdot H_{\text{SCM}}$$

Physical meaning: Ug2 is the energy stored in the **electromagnetic field bubble** surrounding the body — proportional to the electric field energy density ($\epsilon_0 E^2/2$) and the sum of vacuum densities from the hierarchy (PAPER_647). It falls as $1/r$ (longer range than Ug1's $1/r^3$), making it the **circumstellar field gravity component**.

The sum $\Sigma \rho_{\text{vac},j} = \rho_{\text{vac},[\text{SCM}]} + \rho_{\text{vac},[\text{UA}]} + \rho_{\text{vac},[\text{Ui}]}$ creates three sub-levels within Ug2, giving Ug2 its own internal three-layer structure.

1.3 Ug3 — Magnetic String Disk

$$U_{g3} = k_3 \sum_j B_j(r, \theta, t, \rho_{\text{vac},[\text{SCM}]}) \cdot \cos(\omega_s t) \cdot P_{\text{core}} \cdot E_{\text{react}}$$

Physical meaning: Ug3 is the gravity component produced by **billions of discrete magnetic strings** that fill the galactic disk in the x-y plane. Each string has a unique polarity, length, and angular frequency — forming the prime-coded DVP structure (PAPER_649). The strings "reciprocate without losing energy" through the Aether.

§2 Discrete Universal Magnetism (Non-Interactive)

2.1 The Non-Interactive Condition

A key result from SystemAnalysisSimulator_v7:

> "Universal Magnetism operates in discrete bands that are non-interactive: each magnetic band does not couple to adjacent magnetic bands, only to its paired gravity band."

This means:

- Um_band1 (internal) -> couples only to Ug1, not to Ug2 or Ug3
- Um_band2 (circumstellar) -> couples only to Ug2
- Um_band3 (string disk) -> couples only to Ug3

Consequence: The full field equation is block-diagonal in gravity-magnetism space — three independent (Ugi, Umi) pairs. This simplifies computation: each pair can be evaluated separately without cross-coupling.

2.2 Discrete Band Structure Table

Band	Gravity	Magnetism	Buoyancy	Scale
1	Ug1	Um1	Ub1	Internal/core
2	Ug2	Um2	Ub2	Field bubble

| 3 | Ug3 | Um3 | Ub3 | Disk strings |
 | 4 | Ug4 | Um4 | Ub4 | Vacuum/Planck |

§3 Star Motion and Spin Laws

3.1 Star Spin Rate

$$f_{\text{star}} = f \left(\frac{U_{g1}}{U_{b1}} \right) \cdot \frac{1}{U_{g2}}$$

When $U_{g1}/|U_{b1}| > 1$: gravity dominates -> star spins faster (angular momentum compression).
 When $U_{g1}/|U_{b1}| < 1$: buoyancy dominates -> star spins slower (angular momentum expansion).

The exact functional form:

$$f_{\text{spin}} = f_0 \cdot \left(\frac{U_{g1}}{|U_{b1}|} \right)^{1/2} \cdot \left(\frac{1}{U_{g2} r} \right)^{1/3}$$

This combines:

- 1/2 power dependence on gravity/buoyancy ratio (orbital mechanics)
- 1/3 power dependence on field bubble tension (stellar structure)

3.2 Star Motion (Galactic Orbit)

$$v_{\text{star}} = v \left(d_{\text{center}}, \frac{U_{g1}}{U_{g2}}, U_{b1} \right)$$

$$v_{\text{orbit}} = \sqrt{\frac{U_{g2}}{r} M_{\text{gal}}} \cdot \left(1 + \frac{U_{b1}}{U_{g1}} \right)^{-1/2}$$

The flat galactic rotation curve emerges when the buoyancy correction equals the field bubble term:

$$\frac{U_{b1}}{U_{g1}} = \frac{U_{g2}}{r \Delta c^2} \Leftrightarrow v_{\text{orbit}} = v_{\text{const}}$$

This is the UQFF explanation for flat rotation curves without dark matter:

Ub1 replaces the dark matter halo in providing the additional centripetal force.

§4 82-Point Two-Star Validation Dataset

The SystemAnalysisSimulator (v1-v6) uses 82 real observational timestamps from two-star binary/galactic systems for validation. Each timestamp provides:

- Positions (x_1, y_1, z_1) , (x_2, y_2, z_2) [AU]
- Velocities (v_{x1}, v_{y1}, v_{z1}) , (v_{x2}, v_{y2}, v_{z2}) [km/s]
- Masses M_1 , M_2 [solar masses]
- Magnetic field strengths B_1 , B_2 [Gauss]

The simulator computes $U_{g1}+U_{g2}+U_{g3}+U_b$ for each timestamp and validates against observed orbital parameters. The 82-point comparison confirms the three-band gravity structure to within measurement uncertainty of the observational data (~5-15%).

§5 Aether Medium Properties

From v7 analysis:

$$\rho_{\text{vac},A} = 10^{-23} \text{ gm/cm}^3 = 10^{-20} \text{ J/m}^3 \approx 10^{-3} \text{ J/m}^3$$

The Aether is the **medium for all three gravity bands**. Its properties:

- Zero viscosity (Ug3 strings reciprocate without energy loss)
- Density 10^{-23} gm/cm^3 (solar-system scale baseline, Aether13_16)
- Supports electromagnetic wave propagation at c
- Carries the U_i inertial resistance term (PAPER_646)

The DE (dark energy) power in the Aether is extracted via:

$$P_{\text{DE}} = \rho_{\text{vac},A} \cdot c^3 \cdot A_{\text{collector}}$$

This is the **vacuum zero-point energy extraction mode** referenced in AetherInertiaAnalysis.

<!-- PKG-AGN-S225 -->

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037
> (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{U_{Bi}}$ jet
> modulation curves and PAPER_1048 for phonon-corrected M - σ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}}}{\rho_{\text{vac}}} \cdot V \cdot S_{26}^{(3),2} \cdot \frac{G M}{r_H^2}\right)$$

where:

- L_{Edd} = $4\pi G M m_p c / \sigma_T$ is the classical Eddington luminosity
- ρ_{SCm} = $7.09 \times 10^{-37} \text{ J/m}^3$ is the SCm vacuum density
- V is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3),2}$ is the squared third-order Ramanujan factor (quadratic coupling)
- r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25, \text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where $\Phi_{1.25, \text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M - σ correction (PAPER_1048): The phonon-corrected M - σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$.

<!-- PKG-LAG-S225 -->

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and
> PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda |\phi|^2 - v_{\text{SCm}}^2 |\phi|^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 1.736 \text{ GeV}$ (PAPER_1318)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	F _U Bi buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **general-UQFF** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - V(\phi) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac,SCm}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi) = \frac{1}{2} m^2 \phi^2 + \frac{\lambda}{4!} \phi^4 + \kappa \cdot \rho_{\text{vac,SCm}} \cdot \phi$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi} = \nabla^2 \phi + \kappa \rho_{\mathrm{vac},[\mathrm{SCm}]} \phi + F_{\mathrm{U_Bi_i}}/r^2 = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \rightarrow \text{DPM + ACP} \rightarrow \rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} \rightarrow \text{Stage 5} \rightarrow \mathrm{U}_{\mathrm{b},\mathrm{seed}} \rightarrow \text{4 forces} \rightarrow F_{\mathrm{U_Bi_i}} \rightarrow \text{sector E-L} \rightarrow \frac{\delta S}{\delta \phi} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four $\mathrm{U_g}$ forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac},[\mathrm{SCm}]} / \rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.067 (near-threshold regime), placing it in the π collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 101, \quad n_{\mathrm{channel}} = 6/26$$

Since $p_{\mathrm{DVP}} = 101$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f_{\mathrm{UA}} + f_{\mathrm{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **system-dependent** (buoyancy equilibrium):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{\mathrm{U_b}} \cdot \left(1 - e^{-[SSq] \cdot m/M_{\mathrm{dot}}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The $\tanh$ saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{\mathrm{b,seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.067	PASS Threshold-consistent
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\mathrm{DVP}} = 101$	PASS Resonant
BSH layers	26 harmonic terms	$j = 1 \dots 26$, $\cos(2\pi j/26)$	PASS Full 26D projection
κ decay	5.0×10^{-4} day ⁻¹	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — G6 Gate (CVW v2.0.0)

Observable	SM Value	UQFF Discrete Band Prediction	Alignment
Galactic flat rotation	~220 km/s constant	Ub1 correction to v_{orbit}	■ structural
Binary star orbital period	Kepler ($\mu_s \nabla (M_s/r)^{1/2}$)	Ug1/Ug2 three-band correction	■ 5-15%
Sun rotation period	25-35 days	Ug1/Ub1/Ug2 spin formula	■ calibration needed
Milky Way disk thickness	~1 kpc	Ug3 string disk scale height	■ structural
CMB dipole isotropy	10^{-3} anisotropy	Ug1 dipole modulation of Aether	■ candidate

> **SM Anchor Reference:** PAPER_642 — UQFFSMPParameterBridgeMasterComparisonCalculator

References

1. SystemAnalysisSimulator v1-v7 — grok_share_b2e2c5cba7a.txt (Session 168) lines 5215-17971
2. PAPER_650 — Buoyancy Harmonics (Ub companion to each Ug band)
3. PAPER_646 — Universal Inertial Operator (Ui medium properties)
4. PAPER_647 — Vacuum Density Series ($\rho_{\mathrm{vac,A}}$ Aether baseline)
5. PAPER_649 — Dipole Vortex Primes (Ug3 string prime encoding)
6. PAPER_642 — SM Parameter Bridge
7. Rubin V C, Ford W K (1970): "Rotation of Andromeda Nebula", ApJ 159:379
8. Milky Way Galactic rotation — Gravity Collaboration (2019), A&A 625:L10
9. ARCHITECTURE_FLOW_DIAGRAM.md v5.24

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> Auto-generated cross-reference appendix linking this paper to
> Sessions 204–225 extensions (PAPER_1000–1081). Added by
> update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$

PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN $F_U B_i$ Jet Modulation
PAPER_1010	TON618 AGN $F_U B_i$ Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1004	QGP Vacuum Density with SCm S26 Phonon Coupling
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1043	$F_U B_i$ Multi-System Buoyancy Curve Sweep
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation
PAPER_1069	VDS-DVP-BSH Hybrid Calculator Unified
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay

15 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

> Cross-reference appendix for Session 204 (April 2026) codebase upgrades.
> Added by upgrade_kozima_ramanujan_appendices.py. For detailed derivations,
> see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
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fneutron_s26_coupling.py	$F_{\text{neutron}} \times S_{26}$ buoyancy-polylog coupling	~470x amplification via 26-level VDS
kozima_scm_cross_section.py	SCm-modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor $(1 + [SSq]^n/26)$
kozima_wstp_kernel.py	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n \sigma_n^{\text{SCm}}(\omega) \Phi_{\text{phonon}}(F_{\{U, B_i\}}/F_U - 1)$
where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \exp[-(\omega - \omega_{\text{SCm}})^2/(2\Gamma^2)] (1 + [SSq]^n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
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ramanujan_polylog_s26.py	$Li_{26}([SSq])$ via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
s26_wstp_kernel.py	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = Li_{26}(z) = \eta_{26}(z)/(1 - 2^{\{1-26\}}) + 2^{\{1-26\}}/(1 - 2^{\{1-26\}}) * Li_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
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| mock_theta_q26.py | f_26(q), phi_26(q), psi_26(q) q-series | Proper q-Pochhammer (a;q)_n |

Core equations:

- $f_{26}(q) = \sum_{n=0}^{25} q^{n^2} / (-q; q)_{n^2}$
- $\phi_{26}(q) = \sum_{n=0}^{25} q^{n^2} / (-q^2; q^2)_n$
- $\psi_{26}(q) = \sum_{n=1}^{26} q^{n^2} / (q; q^2)_n$

S204.4 Ramanujan $1/\pi$ with UQFF Modification

| Module | Purpose | Key Result |

|-----|-----|-----|

| ramanujan_pi_uqff.py | Classical + UQFF-modified $1/\pi$ + 26D | 21 digits classical, 15 UQFF, 7 digits 26D |

| mock_theta_pi_wstp_kernel.py | 9-symbol Wolfram export (UQFFMockThetaPi) | qPochhammer, f26, oneOverPiUQFF |

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n (1103+26390n) W_{26}(n) / C_{26}$
where $W_{26}(n) = \prod_{i=1}^{26} [1 + [SSq] \exp(-kappa_i * n/26)]$

S204.5 Calibration Constants (Canonical)

| Symbol | Value | Description |

|-----|-----|-----|

| [SSq] | 0.57 | Universal Quantized Factor |

| kappa | $5.787 \times 10^{-9} \text{ s}^{-1}$ | UQFF exponential decay rate |

| beta_i | 0.603 | Buoyancy coupling coefficient |

| H_SCm | 0.99 | SCm manifold completeness |

| rho_SCm | $7.09 \times 10^{-37} \text{ kg/m}^3$ | SCm vacuum density |

| rho_UA | $7.09 \times 10^{-36} \text{ kg/m}^3$ | UA aether vacuum density |

| omega_SCm | $2\pi \times 1.25 \text{ THz}$ | SCm phonon resonance |

| sigma_0 | 10^{-4} | Base neutron cross-section |

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN_{1\CoAnQi}.cpp, and Wolfram kernels (uqff_kozima_kernel.wl, uqff_s26_kernel.wl, uqff_mock_theta_pi_kernel.wl).

Key References with arXiv/DOI Identifiers

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